

Animarium: an open, reproducible pipeline for synthetic populations of Italian cities — from ISTAT sources to open data

Mirko Degli Esposti^{a,*}

^a*Department of Physics and Astronomy (DIFA), University of Bologna, Bologna, Italy*

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ABSTRACT

Local public decisions — siting a service, targeting a campaign, assessing who is exposed to a hazard — are taken at a geography that official statistics do not describe: rich aggregates are published for small areas, rich variables for large ones, never both at once. Spatial microsimulation bridges the gap by reweighting a survey sample against administrative benchmarks, an approach bounded by the sample's support, its access conditions and the municipal level of the benchmarks. We present Animarium, an openly released system built the other way round: its demographic core is generated without a donor population, and every input, including the public-use microdata two later layers draw on, is published data. Synthetic populations are generated from published aggregates alone — census and register tables, census-section counts, the national address register, public-use survey microdata and municipal open data, every source certified in a registry with licence, fingerprint and declared limits of use. A maximum-entropy joint model gives each demographic attribute a declared place; survey respondents donate whole attitudinal vectors; the country of citizenship of foreign residents is conditioned at whatever sub-municipal resolution a municipality publishes, on a declared four-level scale; individuals are placed to census section, year of age and address; households satisfy the census size distribution section by section. The release covers 244 of the 330 municipalities of one Italian region — 95.1% of its residents — plus one city of a neighbouring region: 4,433,976 individuals in 2,051,901 households at the reference date of 1 January 2024.

1. Introduction

Local public decisions are taken at a geography that official statistics do not describe. Where to site a service, whom a communication campaign will reach, which streets a remediation plan should address, who is exposed to a contaminated site: each of these questions is about a joint distribution — who lives where, with whom, in what condition — at a scale well below the level at which counts are published. The finer the decision, the coarser the available evidence. Statistical offices publish rich aggregates for small areas and rich variables for large ones, but never both at once, and the analyst who needs individuals is left to reconstruct them.

Simulation makes the same demand more sharply. Agent-based models, and more recently simulations whose agents are driven by large language models, begin by instantiating a population of individuals, and are only as credible as that population is. An agent needs a place to be: neighbourhood, exposure, proximity, the composition of the street and of the household are properties that do not exist at all in a population resolved at the municipality, however well its municipal margins are reproduced. That place is increasingly being built. Considerable effort now goes into geometric and physical models of cities — building stock in several dimensions, street networks, vegetation, buildings, thermal and hydrological behaviour — assembled under the heading of urban digital twins (Batty, 2018). What these models consistently lack is their inhabitants: the concept descends

from the digital representation of physical assets, and the social component of the city has remained underrepresented in it (Marçal Russo et al., 2026). A synthetic population resolved to the census section and organised into households is, on this reading, the missing half — the residents such a model needs, at the scale and in the place where its geometry already is.

A synthetic population is the standard answer to both demands — a set of simulated individuals whose collective statistics match what is known — and the question is how to build one, and at what geography.

The dominant approach reweights a survey sample against administrative totals, area by area (Tanton, 2014; Hermes and Poulsen, 2012); for Italy the state of the art is the systematic comparison of Scarpa et al. (2026), on precisely the region studied here. That route carries three structural constraints, discussed in Section 2: its support is the donor sample, its richest inputs are restricted microdata, and its benchmarks stop at the municipality, the household layer being the first casualty.

This paper takes the complementary route (Lenormand and Deffuant, 2013). Its demographic core is built without a donor population at all: individuals are drawn from an explicit joint distribution — the maximum-entropy distribution consistent with published aggregate constraints — which by construction puts mass on every combination the constraints allow and, among the distributions compatible with them, is the one that assumes nothing further. Two later layers do use survey microdata, but only the files a statistical office releases for public use, and only as whole donated records: the system as a whole requires nothing that is not public, and its core requires nothing individual at all.

*Corresponding author.

✉ mirko.degliestposti@unibo.it (M. Degli Esposti)
ORCID(s): 0000-0003-0316-3449 (M. Degli Esposti)

On that basis we built, and here present, a complete system: *Animarium*, an openly released set of synthetic populations for 245 Italian municipalities — every municipality of Emilia-Romagna above 3 000 inhabitants, 244 of the region's 330 and 95.1% of its residents, plus Brescia in Lombardy — for a total of 4,433,976 individuals in 2,051,901 households at the reference date of 1 January 2024, generated from public data alone and released, together with the pipeline that produced them and a viewer through which they can be examined; twelve of them carry a sub-municipal articulation and are documented case by case in this paper: the region's provincial capitals, two smaller municipalities, and the city outside the region. The remaining 233 have no articulation below the census section; the same pipeline ran unchanged on all of them, and Section 6 reports their diagnostics as a distribution. The design target throughout is a population fine-grained enough to be *instantiated* rather than merely tabulated: each individual occupies a census section of a few dozen to a few hundred residents, at a street address, in a household whose size the census fixes section by section. Each attribute, moreover, carries the resolution of the source that determined it, from national survey structure down to the section itself where a municipality's open data allow — a graded scheme that Section 4 develops on the country of citizenship of foreign residents, and that is declared rather than hidden. The contribution is the system as a whole, and it rests on five properties that we state here and substantiate in the body of the paper.

1. *Public sources only, to census-section geography.* Every input is a published source — aggregate tables for the demographic core, public-use microdata for the attitudinal and household layers — and the output resolves to the census section, with households constrained by the census size distribution in every section and, where municipal open data permit, individual attributes conditioned at sub-municipal resolution (Sections 3 and 4).
2. *Certified provenance.* All thirty-seven sources are recorded in a registry declaring, for each, its universe, licence, cryptographic fingerprint and limits of use, verifiable by a single command (Section 3).
3. *Reproducibility, and replication.* Regenerating the release from the tagged commit reproduces every file identically; the seed is a declared parameter, so the same command yields either that population or a different draw from the same fitted distribution (Section 6).
4. *Privacy by construction.* The populations are simulated from aggregates, not anonymised from personal records: no released component derives from an identifiable individual, and the public release regime is enforced in the data themselves (Section 5).
5. *A release, not only a method.* The populations are downloadable as an open dataset and browsable in a dependency-free viewer in which every number carries its comparison against the city and against the census, and every view is a citable URL (Section 7).

Two boundaries define what this paper is not. It is not a methods paper: the maximum-entropy solver, its convergence behaviour and its benchmarks are the subject of a separate publication by (Degli Esposti, 2026c) and are used here as an instrument. And although simulation is the motivation stated above, no agent-based or language-model experiment is run or assessed here; what such simulations validly show is a live question (Larooij and Törnberg, 2025) (see also Degli Esposti (2026b)). The full technical record of the system is available as a technical report by (Degli Esposti, 2026a). The paper proceeds by positioning the work among existing approaches (Section 2), then describing the sources and their certification (Section 3), the architecture (Section 4), what in a released record is actually real (Section 5), the measurements at the release tag (Section 6), the viewer and release regimes (Section 7), what the resolution changes for a decision (Section 8), and what the release enables next (Section 9).

2. Related work

Methods for synthetic population generation divide on a question that is easy to state and consequential in practice: does the method require a sample of individual records? Reviews organise the field in various ways (Hermes and Poulsen, 2012; Tanton, 2014; Chapuis et al., 2022), but this is the distinction that governs what can be built, where, and by whom. The sample-based side — reweighting and, more recently, deep generative modelling — learns from microdata; the sample-free side generates from published aggregates alone. This work sits on the second side where it matters most — the demographic core is reconstructed without a donor population — and borrows from the first, for two later layers, the practice of copying whole records from public-use files.

The reweighting family calibrates a survey sample to small-area totals, deterministically by iterative proportional fitting (Deming and Stephan, 1940) or generalised regression (Deville and Särndal, 1992), or probabilistically by combinatorial optimisation, which selects integer copies of respondents directly (Williamson et al., 1998); the trade-offs are mapped by the reviews (Hermes and Poulsen, 2012; Tanton, 2014) and embodied in mature software (Templ et al., 2017). Scarpa et al. (2026) give the sharpest recent instance, and the closest work to ours in territory and intent: reweighting EU-SILC to all 330 municipalities of Emilia-Romagna, validated externally against fiscal records — an exercise that exposes how strongly the result depends on which benchmarks carry the target variable. Their age distributions for large cities are worth a second look: they show spurious peaks, each one a respondent whose calibrated weight was large enough to be copied many times over. The mechanism generalises. A reweighted population is made of copies of a fixed pool, so the more constraints it must satisfy, the more unequal the weights become and the fewer distinct people remain behind a given number of records. Counted as an effective sample size, the distinct donors can fall from

thousands to tens as constraints accumulate (Degli Esposti, 2026c). This is not a defect of execution but what replicating a fixed pool under tightening margins must do.

A newer branch of the same side replaces reweighting with learning. Variational autoencoders were introduced to population synthesis to generate individuals by decoding samples from a learned latent space, recovering attribute combinations that a finite sample never contains (Borysov et al., 2019); normalising flows and, more recently, diffusion models extend the idea to the multimodal categorical distributions that realistic attribute sets produce (Kim and Bansal, 2023; Tang et al., 2025). On the diversity problem that reweighting cannot escape, these methods are a genuine advance: they generate rather than replicate. But they learn from individual-level records, and inherit the other two constraints unchanged. Where the microdata are restricted — as the richest Italian survey files are — whoever holds the aggregates cannot apply the method, nor redistribute its output under the licence of the published tables; and a sample carrying no sub-municipal identifier cannot teach a model to place anyone in a census section. The requirement is definitional rather than incidental: a model that learns a distribution over individuals needs individuals to learn from.

The second family dispenses with the sample. Synthetic reconstruction builds the population from published tables directly, and the sample-free route has been compared with the sample-based one at least since Lenormand and Deffuant (2013), who found it competitive at the household level precisely where sparse samples fail. Its principled form is maximum entropy: among all joint distributions consistent with the published constraints, take the one that assumes nothing else, and draw the population from it. What has historically confined this route to small attribute spaces is the cost of fitting; persistent contrastive divergence removes that barrier at the state-space sizes real municipalities require, with accuracy on constrained margins matching exact solvers and diversity measurably at the maximum the constraints permit (Degli Esposti, 2026c). That methods contribution is not the subject of this paper; the subject is the system built on it — sources, architecture, households, geography, release — and its subject is everything the solver does not decide.

A distinct, recent strand publishes openly released national synthetic populations as contributions in their own right: Canada (Prédhumeau and Manley, 2023); the United States, both a national individual-and-household dataset (Rineer et al., 2025) and a geographically explicit population with social networks (Jiang et al., 2024); Ireland, with four generation methods compared at electoral-division level (Caulfield Curley et al., 2025). These releases share open inputs and a documented generation procedure. Relative to this genre, the release described here differs on four points that the rest of the paper substantiates: geography to the census section, with households constrained by the census size distribution in every section; a source registry that makes provenance checkable rather than narrated; byte-level reproducibility of the entire release from a tagged commit; and an inspection instrument — none of the cited releases

ships a viewer, and we regard the ability of any reader to confront every number with its census counterpart as part of the release, not an accessory to it.

Italy is absent from this genre, and not for want of raw material: its statistical office publishes sub-municipal aggregates of unusual richness (ISTAT, 2026).

Two Italian efforts bound the gap from either side. The FOSSR research-infrastructure cluster, funded under the national recovery plan, offers among its services a Synthetic Populations Generator addressed — in terms close to our own — to research settings in which population microdata are unavailable and information exists only in aggregate form; its code and interface are announced for release through the cluster's virtual research environment (FOSSR, 2026). It is a generator offered as a service, and a welcome one; it is not a released population. At the other end, Garrone (2026) builds a synthetic population of a 746-inhabitant mountain municipality, with residents placed on road-network nodes, to drive an agent-based model of care accessibility: Italian applications are already asking for located synthetic individuals, and currently build them per study, with the resulting data available on request rather than released. What is missing between the two is what this paper contributes: populations that are generated, measured, documented and published, so that the next study begins from a dataset rather than from a pipeline.

Finally, a fast-growing literature drives synthetic populations with large-language-model agents, and a critical strand asks what such simulations validly show (Larooij and Törnberg, 2025). Freely generated persona sets risk collapsing toward an average individual — the diversity failure of the first branch, arrived at by another route — which is one reason statistically controlled populations matter here. What validating that use would require is developed by Degli Esposti (2026b).

3. Sources and their certification

Every municipality receives its data through three doors, at three granularities. The first is the statistical office's dissemination API, which serves census and register tables at the level of the municipality as a whole; these tables measure structure rather than size — a metropolis of 1.4 million and a town of 49 thousand produce the same grid of rows, with only the counts changing. The second is the territorial bases: one workbook per region, one row per census section, carrying population by age and sex, citizenship, education, employment, migratory background and household sizes — the door through which the sub-municipal geography of Sections 4 and 6 enters. The third is municipal open data — chiefly citizenship by country at sub-municipal level — and it is the only optional door: the pipeline is designed to degrade gracefully when a municipality publishes nothing, falling back to what the census margin alone supports, so that the absence of a local source changes the resolution of one attribute, never the feasibility of the population.

What turns this collection of downloads into a method is the registry. All thirty-seven external sources are recorded in a single file that declares, for each: its *universe* (what population, at what date, under which definitions the numbers count), its licence, its access date, a cryptographic fingerprint of the raw files, and its declared limits of use — what the source can and cannot support. The registry is executable rather than narrative: one command re-verifies every fingerprint and every declared count against the files on disk. Sources used only for validation are registered and fingerprinted like the others, but never redistributed.

The universe of a source includes its date, and dates are the one dimension on which the sources cannot be made to agree. A released population is dated by its spine — the population register at 1 January — while its other layers come from other instants: the census of the preceding 31 December, a survey pool of the two preceding years, a section geometry from an earlier census round. Combined as levels these vintages would be incoherent. Combined as conditional form over a single dated spine (Section 4), each contributes structure rather than counts, and the registry records the instant each one came from instead of absorbing it into the population's date.

The reference year is a parameter of the generation rather than an assumption compiled into it, and the acquisition is already plural: each source table is downloaded as a series of seven annual instants, so producing another year is a re-run rather than a new request. Two things bound the choice. The tables do not all cover the same years — some are published for a single one — so the feasible instants are their intersection; and a change of section geometry between census rounds breaks comparability across it.

The source families are eight: census and register tables from the dissemination API, the census-section counts, the national address register, public-use survey microdata, municipal open data, two classification and detail tables from the 2011 census, and open onomastic repertoires. All but the last are ISTAT products under CC-BY-4.0; the repertoires are CC-BY-SA or CC0, with provenance declared per name. Each is recorded individually in the registry, with its universe, access date, fingerprint and declared limits of use, and the registry ships with the release.

Two entries carry the legal weight of the release. The address register is a European *high-value dataset*, released in open form under the Open Data Directive with the prior favourable opinion of the Italian data protection authority: the one source that ties synthetic individuals to real addresses has been vetted by the authority competent to judge it — and Section 5 withholds the address anyway. The survey microdata are *public-use files*, whose disclosure protection is applied at source and inherited here. The attribution file shipped with the data is generated from the registry, so that it cannot drift from the sources actually used.

4. Architecture: the four rings

The architecture answers one question for every attribute of every record: *where does this value come from, and what constrains it?* The answer is organised in four rings (Figure 1). Ring 1 draws the joint demographic core from a maximum-entropy model under census constraints. Ring 2 attaches the attributes no census table can constrain: attitudes and health, donated as whole survey response vectors, and the country of citizenship of foreign residents, whose geographic structure comes — where a municipality publishes it — from municipal open data. Ring 3 places each individual in a census section, a year of age and an address. Ring 4 assembles households under the census size distribution of each section. The rings are strata of conditioning rather than phases of a program: each conditions only on what the rings above it have fixed, so the provenance table of Section 5 can be read off the architecture. Execution follows the strata except in one case — where a municipal source resolves the country below the zone of ring 1, the attribute is finalised together with placement — and there the stratum is unchanged, only the order of computation is.

What the four rings have in common is the discipline announced in Section 1: an attribute is attached at the resolution of the source that determines it, and that resolution is declared rather than levelled. A single record therefore carries several resolutions at once — the difference between a population that reproduces municipal margins and one in which a neighbourhood has a composition. Everything downstream of the four rings — detailed titles, sector label, names, biographies — is a declared deterministic derivation: it adds the legibility that makes a record usable as a persona and no information, which is why the public release regime can drop it without analytic loss (Section 5).

4.1. Ring 1: the maximum-entropy demographic core

Ring 1 models the joint distribution of up to nine categorical attributes — sex, age class, marital status, citizenship class, education, professional condition, migratory background, parents' origin, and sub-municipal zone — as the maximum-entropy distribution consistent with a structured set of constraints. Two design decisions give the constraint set its shape.

The first is the *spine*. Among the available sources, exactly one is treated as hard: the register counts of population by sex, age class and marital status, which define the accounting identity of the municipality — how many people, of which kind, the totals must honour to the unit. Every other source enters *soft*, and enters as *form rather than level*: a census or local table contributes the conditional structure it observes — how education distributes within an age–sex group, how a zone's composition deviates from the city's — rescaled so that its margins agree with the spine. Sources of different universes and reference dates therefore never fight over totals; they contribute shape, and the spine contributes level.

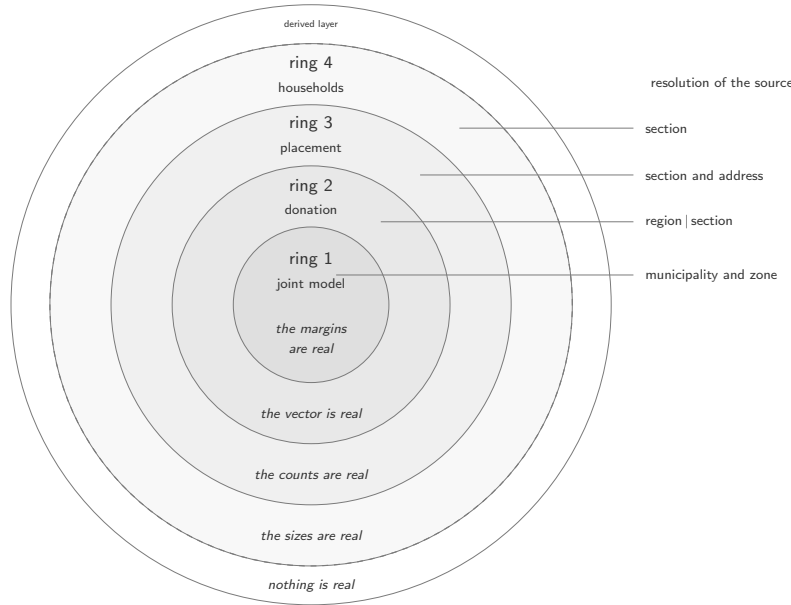


Figure 1: The four rings. Each conditions only on what the rings inside it have fixed, which is why the provenance of a released record can be read off the architecture: the upper half of each band names the ring, the lower half states what, of the attributes it adds, is real (Section 5), and the right-hand column gives the resolution of the source that determines them. Ring 1 draws the demographic core from a maximum-entropy model whose levels come from the population register and whose form from the census; ring 2 donates whole survey response vectors and, for foreign residents, the country of citizenship at whatever sub-municipal resolution a municipality publishes; ring 3 places each individual in a census section, a year of age and an address; ring 4 assembles households under the census size distribution of each section. Outside the dashed boundary, the derived layer — detailed titles, sector, names, biographies — is a deterministic function of what the rings have already generated and adds no information. The shading fades outward with how much of each layer is real.

One word there needs its scope stated. *Hard* describes how a source enters the fit: the spine is imposed as an equality on the fitted distribution, and no other table may move the totals it sets. It does not describe the released population, which is drawn from that distribution and whose cell counts therefore fluctuate around the spine’s expectations by the sampling error of a draw of N individuals — only N itself is exact. The distinction matters because the sampling error, not the fit residual, is what a user of the data meets, and Section 6 reports it beside the floor a perfectly fitted distribution would still show. The asymmetry is less arbitrary than it looks. Register and census are not independent sources: under the permanent census both are dissemination products of the same underlying register of individuals (ISTAT, 2026), so rescaling the census layer onto the spine is, on the spine’s own margins, close to the identity — a claim measured, and found exact to the unit, in Section 6. What the rule buys is therefore not a correction but a guarantee: the totals of the population come from one dated, hard source, and no combination of soft tables can move them.

The same rule resolves the geography. For municipalities with a sub-municipal articulation, the zone coordinate is constrained by section-derived tables — population by age and sex, citizenship, education, employment, migratory background, aggregated from census sections to zones — each entering as the conditional share of a zone within a demographic group, with the group’s total closed on the

spine. Within ring 1, then, one principle covers both the socio-economic attributes and the geography: everything but the spine is conditioned form. Ring 2 applies the same rule to the country of citizenship, at whatever resolution a municipality’s open data support.

Table 1 lists the blocks of the constraint set and, for each, where its form and its levels come from. The table makes visible what the prose above states once: no source ever contributes its own levels, except the register. The mechanism is three lines of code — a conditional share read from the census, multiplied by the register count of the group it conditions on — applied at three scales: register to census for the municipal blocks, municipality to age class for the blocks the census publishes only by class, and municipality to zone for the territorial blocks, where the same share is read from census sections aggregated to the zone and the register count of the group closes the total. The territorial blocks are built in a fixed order: the first, population by zone, sex and age, becomes the row margin that every subsequent block must honour, so that a zone’s citizenship, education and employment profiles are all consistent with one and the same zone population. The hard/soft asymmetry has, moreover, a reason that is measured rather than argued: the register table is the only source in the whole dissemination that articulates marital status — the census declares the dimension in its data structure and collapses it to a single modality in every table it publishes — so for one axis of the spine there is no alternative to treat as soft.

Table 1

The constraint set of ring 1: municipal blocks C1–C10 and territorial blocks Z1–Z5. “Form” is the conditional structure a block contributes; “levels” is the total it is rescaled to. Only C1 contributes levels. C5 and C6 enter as raw census shares and are rescaled in the solver; C9 and C10 read the employed by sex from the block C4 already built. Z1 is the spatial spine: it is built first and becomes the row margin of Z2–Z4. Zone blocks exist only for articulated municipalities. Two blocks tie the families together: GC crosses citizenship with migratory background at municipal level, and Z5 carries that pair to the zone — the path that Section 6 finds does the work when a territorial block is withheld.

block	content	form from	levels from
C1	sex × single-year age × marital status	—	register, exact counts
C2	sex × single-year age × citizenship	census	register, by sex and single-year age
C3, C4	sex × age class × education / condition	census	register, by sex and age class
C5, C6	education / condition × citizenship	census	none (rescaled in the solver)
C7, C8	sex × migratory background; background × parents' origin	census	citizenship margins of C2
C9, C10	sex × professional position / sector	census 2021	employed by sex, from C4
GC	citizenship × migratory background	census	citizenship margins of C2
Z1	zone × sex × five-year age	census sections	register, single-year age
Z2	zone × sex × macro-age × citizenship	census sections	block C2
Z3	zone × sex × education	census sections	block C3
Z4	zone × sex × employed	census sections	block C4 (capped on Z1)
Z5	zone × sex × migratory background	census sections	block C7, consistent with Z2 through GC

Two facts about the territorial blocks bound what the fleet can do. The statistical office publishes a sub-municipal articulation for 11 of the region's 330 municipalities — 38.9% of its population — so the zone coordinate exists only where the office has drawn it; three further municipalities carry a published articulation of the centre-and-hamlets kind that the release does not use, their smallest zones falling below the size at which a zone crossed with sex and sixteen age classes still holds anyone. Second, the two branches agree: when the municipal tables and the census sections refer to the same census year, the harmonisation of the territorial blocks against the municipal ones is exactly the identity, while a one-year offset costs 0.10–0.18% of the mass (Section 6). The full construction, block by block, is documented in the technical report (Degli Esposti, 2026a).

The second decision is to declare what is observed and what is extrapolated. The published tables constrain some interactions directly (education by age and sex, for instance) and are silent on others; the maximum-entropy solution fills the silent interactions with the least committed joint consistent with what is constrained. This is a feature to be stated, not hidden: which pairwise structure is census-backed and which is model extrapolation is a property of the constraint template, recorded with the release, and surfaced to the reader of the viewer (Section 7). Impossible combinations that the sources do not rule out explicitly — an eight-year-old with a doctorate — are excluded by structural zeros declared in the constraint set.

The resulting model is a discrete exponential family over a state space of 161,280 demographic states per zone — of the order of a million states for an articulated municipality. At that size the space is enumerable, so the dual can be solved with exact gradients rather than by sampling: the solver is exact in that sense, converging to a tolerance of 10^{-8} . What it converges to is the maximum-entropy distribution closest to satisfying the constraint set, which is not the same as

satisfying every target: the published tables do not close on one another to the individual, and the residual $2-5 \times 10^{-4}$ across the release is the reconciliation of that disagreement, not solver imprecision (Section 6). From that distribution, N individuals — N the census population of the municipality — are then drawn independently from the fitted distribution. The consequence matters for how Section 6 should be read: the mean relative error of order $2-5 \times 10^{-4}$ on constrained cells is sampling noise, not fit error, which is why it does not grow with the size of the state space across the order of magnitude the 245 municipalities span.

Exactness is a property of the present configuration rather than of the method: the same model form admits a scalable fit by persistent contrastive divergence, shown on synthetic benchmarks to preserve accuracy and pool diversity well past enumeration (Degli Esposti, 2026c), and the implementation carries that solver as an alternative branch, compared with the exact one by Kullback–Leibler divergence wherever both can be computed.

4.2. Ring 2: attributes at the resolution of their own source

Ring 1 constrains what the census publishes at the level of the municipality and its zones. Ring 2 attaches what it does not, and its two contents are chosen by one criterion: their sources live at a resolution other than the constraint set's. The asymmetries run in opposite directions. Attitudes, health and institutional trust exist only in a national survey, which supports conditioning no finer than the region — coarser than ring 1. The country of citizenship of foreign residents exists in the census only as a municipal margin, but several municipalities publish it by neighbourhood, zone or even census section — finer than ring 1. One ring, two directions, and in both cases the attribute is attached at the resolution its source actually observes, with that resolution declared.

Survey attributes: whole-vector donation. No aggregate source publishes attitudes or self-assessed health at municipal level, so ring 2 attaches them by hot-deck donation from public-use survey microdata: each synthetic individual receives the *complete* response vector — twenty-three variables — of one real respondent drawn from the matching cell of region, sex, macro-age and education, with a declared hierarchy of coarser cells as fallback. Copying the vector whole rather than sampling each variable independently preserves the empirical correlation structure at zero modelling cost, because the joint was produced by a person rather than by an assumption; Section 6 reports the check. Two properties follow, and the release inherits both. The public-use file supports conditioning down to the region and no further, so the geography of attitudes stops there while the geography of demography reaches the census section. And the donor pool is regional and finite — 4,629 respondents for one region of the fleet, 8,111 for the other — so each vector serves between 1.4 and 84.3 synthetic individuals and virtually the whole pool is drawn at least once: donor diversity is saturated by construction and cannot be improved by generating more individuals.

What this layer demonstrates should be stated as narrowly as it holds. It shows that the architecture can carry attributes no aggregate source publishes, attaching them at the resolution their source observes and declaring it. It does not show that those attributes are realistic where the population is finest: two synthetic residents of the same section with the same demographic signature are, in this layer, exchangeable.

Country of citizenship: a graded conditional. The census publishes, per municipality, the foreign population by country and sex, and, per census section, the count of foreign residents — but not their composition. Left there, every section of a city would receive the same national mixture, which is precisely wrong: the composition of the foreign population is one of the most strongly geographic attributes a city has. What varies across municipalities is whether a local source exists to say so, and how finely. Ring 2 therefore places the attribute on a declared four-level scale: tier 0 uses the census margin alone and is the default every new municipality receives; tiers 1 and 2 add a municipal table by neighbourhood or by statistical zone, the first typically listing a few dozen countries without sex, the second richer; tier 3 uses a municipal register extract with one row per resident, resolved at the census section. Of the municipalities documented here, Modena stands at tier 0, Brescia at tier 1, Bologna at tier 2 and Parma at tier 3. In every tier the census count of foreign residents per section is a hard envelope.

The local sources are register extracts, of dates other than the census, so the rule of ring 1 applies unchanged: their *form* is used, not their levels. The local table enters as the seed of a two-margin fit whose margins are both censuary — the municipal distribution by country and sex, and the census count of foreign residents in each geographic unit — and, sharing a total by construction, the two margins

are compatible: where the seed puts mass on every cell the margins require — which the structural zeros of Section 4 are declared not to touch — the fit converges, in 9–88 iterations to a residual of order 10^{-11} on every municipality of the release. Imposing the local counts as well would make the margins incompatible, and by a wide margin: the register of one municipality of the fleet counts 40,090 foreign residents where the census counts 37,478.

Tier 0 is not a degraded path but the tested default: the country is conditioned on sex alone within the municipal census margin, and the geography of foreign residents comes entirely from the section counts. It was exercised end-to-end on Modena, whose portal does publish citizenship by neighbourhood but only as a binary — Italian or foreign — which gives the quantity of foreign residents, already in the census per section, and not the composition a conditional needs. The tier machinery is inert there, as it should be: the tier 0 population differs from the one produced with the conditional switched off entirely by a total variation distance of 0.0023, and 0.029 by neighbourhood — sampling noise. Where a local source exists, the same comparison moves the composition by 0.156–0.197, five to seven times more.

Movement is not yet information, since any relabelling across neighbourhoods would move the composition somewhat, so the higher tiers are measured against a permutation null: the share of foreign residents whose country changes under the geographic conditional, divided by the same share under two independent permutations. Across the tier 1–3 municipalities that excess is 2.08–2.57 at the neighbourhood level.

Two results follow. The first is that *a poor but geographically resolved source carries, by this measure, about as much as a rich one*: a tier 1 table of nineteen countries, without sex and with a 10.8% residual, yields 2.08, against 2.57 for a tier 2 table of 155 countries with sex. What matters is the partition, not the vocabulary — and Section 6 explains why, since the geography of an attribute is not deducible from the other blocks: what a local table buys is not precision but the geography itself. This is encouraging for coverage, because a table of country by neighbourhood, even truncated to the twenty largest nationalities, is something many Italian municipalities already publish.

The second is a negative control. Measured at the *section* level rather than the neighbourhood, the tier 1 and tier 2 municipalities show an excess of 1.01 — nothing at all, correctly, because their sources stop at the zone. Only the tier 3 municipality, resolved to 1,320 sections, shows a real excess below the neighbourhood (1.29). The measure finds geographic information where the sources place it, and reports none where they do not.

One door, one mechanism. The formulation is indifferent to the shape of the source. Bologna supplies an aggregated table by zone, Parma a register extract with one row per resident: two very different objects, handled by the same two-margin fit, differing only in the geography that enters the second margin. It is also indifferent to the attribute. What

the scheme requires of a municipality is a table of *something* by a sub-municipal unit; what it returns is that something conditioned at that unit, with the census margins preserved and the tier declared. Citizenship is where a suitable table happens to exist today across several municipalities — the municipal open-data door of Section 3 is an extension point rather than a special case (Section 9).

4.3. Ring 3: placement — section, age, address

Ring 3 turns a demographic profile into a located one, by three assignments that are three declared conditional assumptions: a census section within the individual's zone, an exact year of age within the age class, and a street address within the section. All three have the same form — the finer coordinate is drawn given the coarser one and given whatever the sources cross with it, and assumed independent of everything else — and each is stated separately because each can be contested on its own.

The section is an exact allocation, not a draw. Section counts from the census are honoured by construction: individuals are apportioned by largest remainder within the cells the sources cross with geography, so the per-section error is the residue of integer apportionment rather than of sampling. Sections of the fleet hold 30–309 residents on average, and the synthetic population misses a section's census count by 0.41–1.57 persons, about one in a hundred; drawing the same individuals at random against the same section probabilities would miss it by roughly 9.6, since each section would then carry the noise of its own draw (Section 6). This is what makes section-level statistics usable at all. The assumption is that education, professional condition and migratory background are independent of the section given the zone, sex, broad age class and citizenship: the census crosses those with geography, and the rest is declared to carry no sub-zone signal.

The exact age is drawn within the class. The register supports a single-year profile at municipal level, and the assumption is that the same profile holds within the age classes of each section, with the open top class resolved by a declared convention. It has two measured costs. The model's age bins do not coincide with the published five-year classes, so re-aggregating to those classes leaves a small residual at one boundar — the *seam* of Table 3 —; and since education is constrained at the level of a coarse age class while the exact age arrives afterwards and independently, a few per cent of individuals carry a qualification below its minimum attainment age (Section 6). Both follow from the order of assignment, and both have declared repairs that do not require regenerating the population.

The address is drawn uniformly among the civic numbers of the section, from the national address register: the address exists, the assignment within the section is arbitrary, and the public release regime withholds it while keeping a coordinate randomised inside the same section (Section 5). Coverage is above 99.6% in every municipality of the fleet; where it falls short, populated sections carrying no georeferenced civic number fall back to the zone, which is a property

of how a municipality has certified its addresses rather than of the pipeline. Individuals in institutional co-residence are placed by their own convention, with no address and the zone centroid as coordinate, and remain identifiable as such.

All three draws are deterministic under the release seed, like every other draw in the pipeline — the property that Section 6 certifies as byte-level reproducibility.

4.4. Ring 4: households under the census size distribution

Ring 4 assembles individuals into households. The binding constraint is the census, which publishes the distribution of households by size per section; the assembly honours it section by section, resolving the open top class and the accounting residue explicitly. What the census does not publish is composition — who lives with whom — and there the architecture repeats the honesty of ring 1: household configurations are drawn from a repertoire built on survey microdata, conditioned on the demography already generated, and declared as model, since no census reference exists for household typology at this level. Individuals whom a section's accounting cannot place in ordinary households are identified as institutional population and carry an empty household identifier: 1.1–2.3% across municipalities, identifiable rather than silently absorbed.

5. What, in a record, is actually real

A released record looks like a person: a name, an age, an address, a household, opinions. Before any measurement, the reader is owed a precise statement of what in that record is real — and the answer is not uniform. Table 2 gives it component by component.

Two rows deserve a word beyond the table. The second is the only place where data from real individuals enter the release, and they enter already protected: public-use files are produced by the statistical office through subsampling and disclosure treatment precisely so that they may circulate without conditions, and the release adds a second layer by donating each vector to tens of synthetic individuals. The last row is rendering, and the point of the word “nothing” is that names and biographies are functions of what the model has already generated: they carry no information that is not already in the rows above.

That protection has a consequence worth following through, because it travels with the data. The statistical office states, in the documentation accompanying its public-use files, that the treatment applied to protect confidentiality may lead analyses of those files to results somewhat different from the published figures. The imprecision is therefore declared at source and inherited here: attitudinal statistics computed on the released populations will not reproduce the official survey estimates exactly, and the reason is not the synthesis but the file the synthesis was given. The donation adds a second layer of the same kind. Each vector is replicated across tens of synthetic individuals — some 1.4 to 84.3 times, depending on municipal population —

Table 2
Provenance of the components of a released record.

component	provenance	what is real
demographic attributes (nine, from sex to zone: see Table 1)	drawn from a maximum-entropy distribution under census constraints	the <i>aggregates</i> are real; the individual is a sample of a distribution, and no record corresponds to any one
attitudes and health (twenty-three survey variables)	the complete response vector of one real survey respondent, copied whole	the vector is real — already protected at source by the statistical office's public-use release, and reused across tens of synthetic individuals, so no combination is unique to one record
census section, exact age, address	allocated within the census section; the address uniformly among its registered civic numbers	the <i>counts</i> per section are real; the assignment carries no information about anyone
household id and role	assembled under the census size distribution per section; composition from a survey-based repertoire	the size distribution is real per section; the household is a model
names, detailed titles, sector label, biography	deterministic rendering from registered repertoires and census-conditioned structure	nothing: plausible by construction, and shared by many — a name individuates no one

so no combination of attitudinal responses in the release is unique to a record; what remains real is one respondent's already-perturbed answers, never their identity and never their geography. Neither step is offered as a proof that disclosure risk is nil, which is not ours to certify; both are stated so that a reader can see exactly which treatments a released value has passed through, and who applied them.

The distinction underneath the table is the one that matters legally. These populations are not anonymised data. Anonymisation starts from real individual records and must demonstrate that the link to the person has been broken, with re-identification as the residual risk; simulation from aggregates starts from tables of counts, there never was a link to break. The one component that derives from real individuals — the donated survey vector — arrives already subject to the statistical office's disclosure treatment and is replicated across tens of synthetic records, so that no

combination in the release is unique to one of them. We do not claim that re-identification risk is formally zero, which is not ours to certify; we claim that the released records are not derived from identified individuals, and we state which treatment each component has passed through, and who applied it.

The public release regime is nonetheless conservative, and enforced in the data rather than in a terms-of-use clause: released files carry no names and no street addresses, and every individual's coordinate is randomised within its census section (median displacement 87 m, 100% of points remaining within their section), so that section-level statistics are preserved exactly while nothing finer can be read from a point.

6. Measurements at the release tag

Each ring of Section 4 makes a guarantee, and this section is the ledger that tests them: for each guarantee, one number, measured at the release tag against a declared comparison — a census table, a null model, a sampling floor. Where a diagnostic revealed a limit, the limit is reported with its mechanism rather than absorbed. The release has two layers and they answer different questions, so the ledger is in two parts: twelve reference municipalities, articulated into sub-municipal zones and measured in full detail (Section 6.2), and the 233 municipalities of the regional fleet, which carry no sub-municipal articulation and are reported in aggregate (Section 6.3).

Reproducibility. Regenerating the entire release from the tagged commit reproduces every released file identically on a single desktop workstation (a 12-core AMD Ryzen AI 9 HX 375 with 64 GB of memory, running Linux under WSL2): byte for byte for all but two files per municipality, which embed the instant of generation and are compared modulo that field. Determinism is not a side effect but a design constraint: every draw in the pipeline is seeded. The claim is verified against the *published* bundle, not a working copy: what is reproduced is what anyone can download. The seed is a parameter rather than a constant, so the same command reproduces this population or draws another from the same fitted distributions — a distinction that matters for simulation, where a single population is one realisation and a study may want several. The pipeline itself was developed with AI assistance; what the release certifies is not how the code was written but that it runs, and that its output is reproducible from the tagged commit by anyone.

The demographic base: register and census agree exactly. The spine of ring 1 comes from the population register; the conditional structure comes from the permanent census. Before trusting the form-not-levels construction, the two sources were compared directly, at the level of decoded source tables, before any processing: sex by single year of age, register at 1 January against census at 31 December of the preceding day, 2,821 cells over fifteen municipalities — zero discrepant cells, with a maximum absolute difference of zero. The identity is no coincidence: under the permanent

census both dissemination products are projections of the same statistical register of individuals (ISTAT, 2026), so the spine and the conditional layer are two views of one underlying accounting — which is precisely what makes treating one as hard and the other as form consistent rather than convenient.

6.1. Four errors, and which question each answers

The acronym MRE has been used for three different quantities, and a fourth is needed where they fail; the difference between them governs every diagnostic below. For a constrained cell c with positive target t_c , the fitted distribution assigns an expected count e_c and the drawn population contains n_c individuals. The *fit residual*, $|e_c - t_c|/t_c$ averaged over positive-target cells, answers whether the model reproduces the margins; it is small by construction — the dual converges to tolerance 10^{-8} , and what remains, $2-5 \times 10^{-4}$ across the release, is the reconciliation of census tables that do not close on each other to the individual rather than solver imprecision. It measures the sources, not the population.

The *sampling error*, $|n_c - t_c|/t_c$ averaged the same way, answers how far the drawn population is from the tables — what a user actually meets — and it is two orders of magnitude larger, 3.95–15.81%: a cell with expected count e_c in a draw of N individuals is binomial, with standard deviation $\sqrt{e_c(1 - p_c)}$ and hence close to $\sqrt{e_c}$ for the cell probabilities at issue, so an *exactly* fitted distribution already produces an error of this size. That is the *floor*, computed cell by cell from the constraint set, and it is what the sampling error has to be read against: an observed error at its floor has nothing left to explain.

The floor is a mean of ratios, and a ratio diverges as its denominator goes to zero, so a cell with a very small positive target can dominate the average. One municipality of the fleet reads a floor of three thousand per cent, and the cause is two cells out of 1 989: foreign men aged 65 and over in one zone, whose targets are of order 10^{-9} — a numerical residue of the harmonisation rather than a count. They account for 99.6% of that municipality's floor; without them it reads 13%, in line with the rest of the fleet. Nothing about that municipality's population is anomalous, and a target that small is arguably a structural zero the constraint set failed to declare.

Where expectations are that small, a scale-free quantity is the one to consult. That is the standardised residual $z_c = (n_c - e_c)/\sqrt{e_c}$, whose mean absolute value has expectation $\sqrt{2/\pi} \approx 0.80$ under independent draws, this being the expected absolute value of a standard normal. The approximation holds while e_c is large enough for the binomial to be near-normal, and Table 3 shows where it stops: every $|z|$ above ten sits on a cell expecting one individual, while the municipality whose worst cell expects five thousand reads $|z|_{\max}$ just above three.

6.2. The reference municipalities

Ring 1: the draw, not the fit. Table 3 reports fit and placement diagnostics for the twelve municipalities that

carry sub-municipal articulation. Observed sampling error is at or below its floor almost everywhere, and the mean $|z|$ stays in 0.78–0.91 against a theoretical 0.80: the drawn populations behave, to first order, as independent samples from their fitted distributions. No declared hard zero is violated, and the impossible demographic combinations excluded by construction (28–35, depending on which cells the local constraint set resolves) were re-verified at population level with zero occurrences.

Where the residuals are not independent, and why. The dispersion of the standardised errors is not unity: it runs 0.98–1.39 across the reference municipalities and grows with the number of constrained cells. The cause is the constraint set rather than the sampling — the cells are overlapping margins of one joint distribution, so their residuals are mechanically correlated, the more so the denser the template — and under that dependence the variance of z exceeds unity while its mean absolute value does not. The consequence is bounded: tail quantiles of z should not be read against a normal reference, and the share of cells beyond $|z| > 3$ (0.00–1.15%, against 0.3% expected) is the quantity to consult instead.

The fit is at its floor block by block, not only on average. A mean over all constrained cells can hide a block that is systematically off, so Table 4 opens the constraint set of one municipality — Cesena, 12 zones, 16 blocks — and compares observed against floor within each. Sampling error varies by more than an order of magnitude across blocks, from well under one per cent on the two-way age blocks to above twenty on the three-way block resolving parents' origin, and it tracks the floor throughout, because both are governed by the expected count per cell. The finest block of all, resolving 384 cells of zone by sex by age by citizenship, reads 11.9% against a floor of 14.6%: the error is largest exactly where the constraint set is most demanding, and still below what an exactly fitted model would produce. No block sits materially above its floor.

Ring 2: what the donation preserves, and what it costs. Copying whole response vectors is meant to preserve the joint structure of the survey, and it does: across the pairs the attitudinal variables admit, the correlations of the synthetic population match those of the donor pool, and the donors' own pattern of missingness is transmitted intact — coverage in the population tracks coverage in the pool to within 2% on every variable. The cost is reuse, and it scales with municipal population: each donor serves 1.4–84.3 synthetic individuals across the release, and in the largest municipalities virtually the whole regional pool is drawn at least once (99.7–100%). Donor diversity is therefore saturated by construction in the large cases, and the one municipality drawing on a different regional pool carries proportionally more distinct donors at equal population, as the design predicts. The effective number of donors behind any subpopulation is computed per variable battery on its own universe and ships with the release — the quantity to consult before reading attitudinal statistics of small groups, and one that differs from the nominal sample size by a factor of five to twenty.

Table 3

Per-municipality diagnostics for the twelve reference municipalities at tag release-v2.0. *Municipality*: population, number of zones (— where none is published) and census sections. *Fit*: constrained cells with positive target; sampling MRE and the floor it must be read against; mean and standard deviation of the per-cell standardised error, whose mean equals $\sqrt{2/\pi} \approx 0.80$ under independent draws; and the largest $|z|$ with, in brackets, the expected count of the cell producing it. *Placement*: mean absolute error of section populations (exact allocation; multinomial baseline ≈ 9.6) and mean absolute *seam* residual per section. All four quantities are defined in Section 6.1, and the two per-section columns scale with mean section size, which the first two columns give.

	municipality			fit						placement	
	pop.	zones	sect.	cells ($\alpha > 0$)	MRE (%)	floor (%)	$\overline{ z }$	sd(z)	max $ z $ (at e_c)	alloc. MAE (per sect.)	seam (per sect.)
Bologna	390 098	18	2 224	1 751	7.05	5.29	0.84	1.391	36.0 (1)	1.36	5.11
Brescia	198 259	33	1 822	2 949	7.82	9.52	0.80	1.012	7.1 (2)	1.57	5.34
Parma	198 121	13	1 357	1 349	4.99	6.10	0.80	1.014	3.7 (9)	1.44	5.74
Modena	184 597	4	2 186	629	3.95	5.07	0.82	1.018	3.1 (5577)	0.74	2.61
Reggio nell'Emilia	171 207	4	1 813	629	4.74	5.32	0.78	0.979	3.8 (27)	0.90	3.32
Ravenna	156 304	10	2 376	1 108	7.99	7.50	0.85	1.164	17.0 (1)	0.74	2.27
Rimini	150 046	6	1 888	789	4.55	5.41	0.82	1.014	3.3 (13)	0.89	2.57
Ferrara	129 391	—	1 761	261	9.33	7.95	0.80	1.098	7.1 (2)	0.72	2.41
Forlì	117 050	21	1 519	1 989	11.04	3 189	0.85	1.092	10.0 (1)	1.18	4.08
Cesena	96 066	12	1 275	1 267	8.19	10.12	0.80	0.996	3.2 (7)	1.05	3.02
Piacenza	102 887	4	1 009	629	5.48	6.70	0.91	1.118	4.1 (166)	1.07	3.93
Castenaso	16 357	—	154	257	15.81	23.68	0.80	1.000	2.9 (170)	0.87	3.21

Table 4

The constraint set of Cesena opened block by block: observed sampling error against its floor, with the mean standardised error and the count of cells beyond $|z| > 3$. Error scales with the fineness of the block and tracks the floor throughout.

constraint block	cells	MRE (%)	floor (%)	mean $ z $	$ z > 3$
sex × background	12	2.14	2.81	1.03	1
age × education	9	0.31	1.27	0.24	0
age × activity	20	0.26	1.31	0.20	0
citizenship × education	12	1.82	1.98	0.82	0
citizenship × activity	14	2.56	2.34	1.13	0
citizenship × background	12	1.39	1.95	1.23	0
zone × sex × age	192	4.13	5.38	0.78	0
zone × sex × education	144	4.41	5.14	0.82	0
zone × sex × background	144	9.65	12.15	0.84	2
sex × age × marital status	52	9.40	8.79	0.80	0
sex × age × citizenship	32	3.33	3.47	0.78	0
sex × age × education	78	3.88	5.31	0.69	0
sex × age × activity	72	13.10	19.04	0.70	0
sex × background × parents' origin	29	22.09	21.91	0.99	1
zone × sex × age × citizenship	384	11.87	14.56	0.80	0
zone × sex × age × activity	96	4.72	6.67	0.75	0

What the model reconstructs, and what it cannot. Every diagnostic so far verifies that the population reproduces what the constraint set imposes. None of them says whether the relations the model fills in by itself — those on which no constraint speaks — resemble the real population, and those are the relations a simulation or a decision would lean on. The release is in an unusual position to answer, because the constraint set contains territorial blocks that can be removed: taking one out, refitting and drawing a population turns the removed block into ground truth, since it comes from the census and was not used.

Two blocks admit the test cleanly. Z3 gives education by zone and sex; Z5 gives migratory background by zone and sex. For each, the reconstruction is read between two references: the floor, the error that remains when the block is constrained, which is sampling noise and is irreducible under

the i.i.d. design; and *independence*, the error of assuming the attribute has no geography at all given sex, which is complete ignorance. Where the hold-out falls between the two says how much of that geography survives through the other blocks.

The answer differs sharply between the two, and the difference is the result (Table 5). One municipality reads slightly above 100%, which is not a paradox: the gap between floor and independence is there only three thousandths wide, and the residual constraints happen to push a shade past it. For education the hold-out sits at 94% of the gap across five municipalities (84–104%): removing Z3, the model reverts almost exactly to independence. For migratory background it sits at 24% (16–38%): three quarters of the geography is recovered without the block. The asymmetry is not a property of the attributes but of the constraint set that connects them.

Table 5

Hold-out of two territorial blocks. Each block is removed from the constraint set, the model refitted with the same pool and the population redrawn; the removed block, which comes from the census, is the ground truth. Figures are total variation distances between the census distribution and the synthetic one. *Floor*: the block is constrained, so the error is sampling noise. *Hold-out*: the block is removed. *Independence*: the attribute is assumed independent of the zone given sex. The last column places the hold-out between the other two. Values are single draws, not averages over replications, so differences of a few per cent between municipalities should not be read as meaningful.

municipality	floor	hold-out	independence	of the gap
<i>Z3 — education by zone and sex</i>				
Bologna	0.0086	0.0713	0.0818	86%
Parma	0.0103	0.0531	0.0545	97%
Modena	0.0063	0.0283	0.0326	84%
Reggio nell'Emilia	0.0068	0.0460	0.0467	98%
Rimini	0.0071	0.0379	0.0368	104%
<i>Z5 — migratory background by zone and sex</i>				
Bologna	0.0065	0.0210	0.0449	38%
Parma	0.0079	0.0173	0.0579	19%
Modena	0.0039	0.0141	0.0445	25%
Reggio nell'Emilia	0.0053	0.0119	0.0455	16%
Rimini	0.0060	0.0119	0.0351	20%

Background reaches the zone through a chain that exists — citizenship is constrained by zone (Z2), and background by citizenship (GC) — while education has no comparable path: the block that ties it to citizenship is municipal and weak. Maximum entropy propagates structure where a chain of constraints carries it, and declares ignorance where none does.

Two consequences follow, and both belong in the paper rather than in a reader's inference. The first is a rule for using the release: before relying on an association between two attributes, one should ask whether the constraint set connects them — the blocks are listed in Table 1, and the answer is legible there. The second concerns the graded scheme of Section 4: if the geography of an attribute is not deducible from the others, then holding the territorial table is not a marginal improvement over not holding it. It is the difference between having that geography and not having it, which is consistent with what Section 4 found by a different measure: what a local table buys is not precision but the geography itself, and a coarse table that has it outperforms no table at all by far more than a rich table outperforms a coarse one.

Ring 4: the friction of the household constraint. The census size distribution per section is honoured, its open top class resolved by a declared convention. On the reference municipality the household total matches the census to the digit and the size distribution deviates by less than 0.8% in every class; of 2,118 sections, 994 close exactly. The rest need an adjustment, and which adjustment is itself informative: the census may overstate a section relative to its population, the open class may need truncating, or the section may hold people who cannot be placed in ordinary households at all. The last case is how the institutional population is identified — 1.1–2.3% of individuals across the fleet, who carry an empty household identifier rather than being silently absorbed into families.

6.3. The regional fleet

The 233 municipalities generated in the second campaign carry no sub-municipal articulation — their census sections are the finest spatial unit and there is no zone layer above them — so what is reported here is not a per-municipality ledger but the distribution of the same diagnostics across a region. Together with the reference cases they cover 244 of the 330 municipalities of Emilia-Romagna (74%), and 4,433,976 individuals.

What holds across all of them. The structural exclusions — demographic combinations declared impossible and removed from the state space before fitting — carry zero probability mass in *every* municipality of the fleet, which turns a property previously verified on a single case into one verified on 233. The generated total matches the census population exactly in every municipality. The fit residual stays within 2.2×10^{-4} – 1.0×10^{-3} with a median of 4.8×10^{-4} , showing no trend with population: the solver does not degrade as the state space grows, which is the property that makes the fleet possible at all.

The admission threshold, verified after the fact. Municipalities were admitted above 3 000 inhabitants, and the reason is a ratio the release makes measurable: individuals per state of the fitted model. Below one, a municipality is a sample smaller than its own model, and no amount of exact fitting can make its finest cells meaningful. Table 6 reports that ratio and three diagnostics by population band: the ratio rises monotonically from 1.30, 2.27, 4.21, 8.64, 20.1, and the relative placement error falls with it across the bands that carry enough municipalities to have a stable median. The threshold sits at the right place with a narrow margin — at 2 500 the lowest band would have fallen below one — and the case that confirms it is a municipality excluded after generation: Sarmato, admitted on a register population above

Table 6

The regional fleet by population band: medians of individuals per model state, of the fit residual, of donor reuse, and of the section-level placement error relative to mean section size. The first column is the admission criterion, verified after generation on 233 municipalities rather than assumed. The highest band contains two municipalities, so its medians are single values.

population band	municipalities	ind./state	fit residual ($\times 10^{-4}$)	donor reuse ($\times$)	rel. MAE (%)
3 000–5 000	51	1.30	4.9	1.6	1.0
5 000–10 000	93	2.27	4.7	2.1	0.7
10 000–20 000	65	4.21	5.0	3.1	0.6
20 000–50 000	22	8.64	4.6	6.1	0.5
above 50 000	2	20.12	5.0	14.3	1.0

the threshold but whose census population of 2,931 gives a ratio of 0.95. Its files remain on disk and out of the release.

Placement without articulation. Section populations are reproduced with a mean absolute error of 0.41–1.57 individuals across the release, on sections averaging 30–309 residents; in relative terms the error is 0.26–2.11% of section size, with a median of 0.68%, and the correlation between observed and synthetic section totals does not fall below 0.9994 in any municipality. The foreign-resident count per section, which is the constraint most exposed to small numbers, is reproduced at a correlation above 0.986 throughout. These are the quantities a user of the fleet should consult: what the fleet guarantees is the section, not the neighbourhood, because the fleet municipalities have no neighbourhood.

7. The Animarium viewer and the release regimes

A data release that can only be downloaded is, in practice, released only to those who already know what to ask of it. Animarium (<https://animarium.it>) is the release's inspection instrument: a viewer in which any reader can open a municipality, filter its population, and see every statistic next to the number it should be compared with. We regard it as part of the release, not an accessory: several of the limits declared in Sections 4 and 6 were surfaced, and are kept honest, by the instrument that displays them. The interface is available in Italian and English.

What the viewer is technically. A single hand-written HTML page: no framework, no build step, no server-side component. The page loads an embedded analytical database compiled to WebAssembly, reads the released Parquet files directly over HTTP range requests — fetching only the byte ranges a query touches, so that opening a municipality does not mean downloading it — and draws its bars, maps and export graphics itself.

The choice is deliberate and load-bearing: a static page served from a folder is the most reproducible artefact a viewer can be, anyone can serve it locally from the open bundle with no installation, and the absence of a dependency graph means that what is served is the released files themselves, byte for byte, rather than a rendering of them. The one external dependency is opt-in: the cartographic base layer is off by default, the panel drawing its choropleths on an empty

background, and a reader who switches a tile provider on is told it is an external service.

Every number carries its comparison. The panel is organised around one idea. Filters are the bars themselves: clicking a bar filters the population, active filters become removable pills, and every marginal on screen then shows up to three markers with three distinct meanings. The *bar* is the filtered subpopulation. The *tick* is the whole city: the gap between bar and tick measures the association between the filter and the displayed attribute. The *diamond* is the census count at the filter's level: the gap between bar and diamond measures model error, in the only place where model error is defined — where a block of the constraint set contains the filter's attributes together with the displayed one. That condition holds for 67 of 333 attribute pairs in the articulated configuration and 26 of 96 in the unarticulated one — a property of the constraint template, identical for every municipality — and where the diamond cannot exist the panel says so instead of hiding it. The boundary of Section 4 between what is census-backed and what is maximum-entropy extrapolation is thereby not a paragraph in this paper but a visible property of every view: the reader who filters by an attribute pair the census never crosses sees, explicitly, that the comparison is not available.

Views are citable. The entire state of the panel — municipality, filters, open views, map mode — lives in the URL query string, so a view can be sent in one line and cited in a paper: the recommended citation is the URL together with the SHA-256 prefix of the municipality's Parquet file, whose per-file hashes are published with the release. The address reproduces the view and the fingerprint says which data it showed, so the citation survives redeployment and detects both tampering and the ordinary case of a file since regenerated. We give addresses rather than screenshots throughout: a reader can open a view and interrogate the population that produced it, which no reproduction of the interface would allow.

Deposit and service are different objects. What this paper measures is the deposited release: a tagged commit and an archived dataset with a persistent identifier, from which every released file is reproduced byte for byte (Section 6). The site is a living service and is meant to grow — further municipalities and regions will appear there before any deposit — and the two are kept honest by keeping them apart, with the file fingerprints as the bridge. A citable address on

a growing service requires a commitment, and we state it: municipality identifiers and the state parameters in the query string are a public interface, later versions read the state written by this one, and an address that cannot be honoured is reported rather than silently resolved to something else. That holds for as long as the service is maintained — deliberately a weaker guarantee than the deposit's, since the address is a convenience and the deposit is the guarantee.

What serving it requires. Static files over HTTPS, range requests and no file-size ceiling: common but not universal, and the deployed folder carries a self-test for range support because that requirement fails silently. The domain is registered independently of the host, so the addresses cited here survive a change of platform; switching the site off would not withdraw what has been downloaded.

Release regimes. The viewer serves the public regime of Section 5, enforced in the files it reads; the narrative layer and the export with exact addresses never enter the public bundle. The viewer cannot leak what the bundle does not contain.

8. What the resolution changes: an illustration

The argument of Section 1 was that local decisions are taken at a geography official statistics do not describe. It is worth showing, on the release itself, what that costs — not because the example settles any planning question, but because the size of the gap is not obvious in advance, and turns out not to be small.

The question. A programme intends to reach the least-served part of a city: the fifth of census sections furthest from a pharmacy. How many residents there are aged 75 or over *and* hold at most lower secondary education? The target is deliberately defined by two attributes at once, because that is where the published tables stop. The census gives, per section, the population by age and sex, and the population by education and sex; it does not give their joint distribution. An analyst can have each margin and still not know how many people are in both categories at the same address.

Three answers. The first, *A*, uses no geography: the municipal share of the target applied to the population of the least-served sections, as though the target were spread like everyone else. The second, *B*, uses the published territorial tables to their limit: in each section, the share of elderly multiplied by the share of low education, that is, the two attributes assumed independent *within* the section. It is the best an analyst can do from public sources today. The third, *C*, counts the target in the released population, where the two attributes sit on the same individual. **A against C measures what ignoring geography costs; B against C measures what having the margins but not the joint costs**, and that is the comparison this release is about.

What the example is not. It is not a study of accessibility. Distances are straight-line, measured from the centroid of each census section — the finest constrained spatial unit. The pharmacies come from OpenStreetMap, chosen because it is one source with one schema for the whole country, at the price of a completeness nobody certifies: against

the municipal register that one of these cities publishes, OpenStreetMap locates 94% of its pharmacies to a median of 9 metres, missing a few in the periphery and adding parapharmacies the register does not list. A missing point does not only change the counts, it can move which sections fall in the least-served fifth; the numbers below measure a difference between three ways of counting, on one fixed set of points, not a distance anyone walks.

The result (Table 7). Ignoring geography errs in both directions: *A* departs from *C* by -7.6% to $+22.8\%$, overstating in eight cities and understating in two, with nothing in the published statistics to say which case a city is in. Using the margins without the joint errs in one direction only, and by more: *B* understates the target in every one of the ten cities, by -17.3% to -26.2% , with a median of -21.2% . A planner who multiplies the two published shares will miss roughly a fifth of the people the programme is meant to reach, in every city of the region.

The sign of that error is the one the world predicts. Within a section, age and low education are positively associated — older cohorts left school earlier — so treating them as independent loses the overlap. The release recovers the association because the constraint set carries it, through the municipal blocks that cross age, education and citizenship (Table 1). That is also the assumption the example rests on, and it should be read with Section 6 in hand: the geography of each attribute is constrained by the census, their joint within a section is maximum entropy conditioned on municipal structure, and the hold-out measures how far such unconstrained structure can be trusted. What this illustration shows is that the joint changes the answer by a fifth, not that it is exact.

9. Possibilities and outlook

The release covers one region above a population threshold, and the first thing to say about it is what it is not: a sample of municipalities chosen for their convenience. Above the threshold it is the region, exhaustively — of 330 municipalities, 244 lie above 3 000 inhabitants and hold 95.1% of the population — so the question this kind of work usually has to answer, why these places and not others, is replaced by one the registry answers: which municipalities fall below the threshold, and why. Below it the sampling floor of Section 6 dominates the constrained cells, a municipality becomes a sample smaller than its own model (Section 6.3), and a synthetic population adds little to the published tables. Brescia, in Lombardy, is the one municipality outside the region: it is included because its open data provided the first tier 1 conditional of Section 4, and it remains the tier's reference case. What makes the collection extensible is that two of the three doors are national: census and register tables arrive through a dissemination API, and census-section counts and the address register are published for the whole country, so the inputs of rings 1 and 3 are available for any Italian municipality without negotiation. Only the survey microdata require a request, made once and valid for all municipalities, and only the sub-municipal

Table 7

Residents aged 75 or over with at most lower secondary education, in the fifth of census sections furthest from a pharmacy, in ten Emilia-Romagna provincial capitals. *A*: the municipal share of the target applied to the population of those sections. *B*: in each section, its elderly share times its low-education share, that is, the published margins with independence assumed within the section. *C*: the count in the released population. Departures are relative to *C*. Pharmacy locations from OpenStreetMap; straight-line distance from section centroids. The comparison is between three ways of counting one population, not between measurements of accessibility.

municipality	count in the least-served fifth			departure from <i>C</i>	
	<i>A</i>	<i>B</i>	<i>C</i>	<i>A</i> (%)	<i>B</i> (%)
Bologna	2 230	1 701	2 060	+8.2	−17.4
Parma	2 131	1 375	1 735	+22.8	−20.8
Modena	1 363	1 104	1 334	+2.2	−17.3
Reggio nell'Emilia	1 441	1 060	1 325	+8.8	−20.0
Ravenna	1 949	1 642	2 095	−7.0	−21.6
Rimini	3 240	2 269	2 967	+9.2	−23.5
Ferrara	1 222	842	1 113	+9.8	−24.3
Forlì	1 708	1 414	1 849	−7.6	−23.5
Cesena	1 446	1 055	1 431	+1.0	−26.2
Piacenza	656	437	546	+20.1	−20.0
median				+8.5	−21.2

conditional depends on what an individual administration chooses to publish. That the architecture travels is not a promise but something the release already exercises: the city outside the region draws on a different regional survey pool — 8,111 respondents against 4,629 — through the same code path, with a registry entry instead of a code change, and its diagnostics fall in the same bands as the rest (Table 3). A second region is in production at the time of writing; what a new region costs is the acquisition of its published tables, not a change to the method.

The same door admits more than it currently carries. Household income is the most consequential absence, and the route to it is charted in two steps. The first is a coherent assignment of economic sector and professional position to each employed individual — the released records carry a sector label derived from the census structure, but not one that is consistent within a person and a household: the constraint set already carries both as municipal margins (C9 and C10 in Table 1), but not yet as attributes consistent within a person and a household, and that is work in progress not reported here. The second is income itself, conditioned on those attributes: survey microdata supply the variable at regional resolution, and the tax authority publishes income class frequencies per municipality — exactly the pairing that Scarpa et al. (2026) show to be decisive, since an attribute not carried by the constraints is reproduced poorly and becomes accurate as soon as a benchmark carries it. In the architecture of Section 4 that is a ring 2 attribute with a municipal margin, not a new method. The statistical office has meanwhile extended its section-level release with country of birth, dwellings and household car ownership, each of which enters as a new block of the constraint set or as a new conditional. And nothing in the tier scheme is specific to citizenship: it requires a table of some attribute by some sub-municipal unit, so any variable an administration

publishes at that resolution can be conditioned on it, with the tier declared as it is for the country.

Section 8 has shown one such decision in miniature; the class is wider. What the release supports, in the end, are decisions taken below the level at which official statistics are published. Institutional risk communication is the case that motivated the work: the residents of the neighbourhoods around a contaminated site are not a homogeneous public, and a campaign designed for the municipal average reaches a composition that does not exist anywhere in particular. The same holds for the siting of a local service, for estimating who is within walking distance of one, and for the ex-ante assessment of a measure whose incidence depends on household composition rather than on individual attributes. In each case the synthetic population is not the analysis but its substrate: it supplies individuals where the sources supply only counts, and it declares, attribute by attribute, how much of what it supplies is observed.

Simulation is the second use, and the one this paper deliberately does not evaluate. Released records carry a deterministic narrative rendering that makes them usable as personas for language-model agents, and work on that layer is under way; what such simulations validly show, and what would have to be measured before believing them, is the subject of Degli Esposti (2026b) and of the critical discussion in Larooij and Törnberg (2025). The claim made here is narrower and prior to it: whatever an agent population is asked to demonstrate, it should begin from individuals whose statistical provenance is known.

One limitation deserves to be stated plainly, because it bounds every use proposed above. A maximum-entropy population is a population *compatible with the information available*, not an identified reconstruction of the true joint distribution. Where the constraint set speaks, the release reproduces it to the sampling floor; where it is silent, the

model fills the gap with the least committed assumption, and Section 6 measures what that costs: for one attribute almost everything, for another almost nothing, depending on whether a chain of constraints connects it to the geography. A study that leans on an unconstrained association is leaning on that assumption, not on the census; the remedy, where it matters, is to add the table rather than to trust the model.

Beyond attributes lies the direction we regard as the natural continuation of this work, and it returns to where Section 1 began. A synthetic individual currently has a place but no knowledge of it: the record states a section, an address and a coordinate, and says nothing about what is around them — which streets, which services within reach, what the neighbourhood looks like from the doorstep. Urban digital twins describe exactly that, and are being asked, from within their own literature, to find a way of representing the people who live in them (Marçal Russo et al., 2026). Coupling the two would give each individual a local knowledge consistent with where they live, derived from the geometry rather than assumed: residents who move through a street network that exists, see a built environment that is measured, and respond to interventions in it. The step is from a population that is *located* to one that is *situated*, and it is what would make these individuals usable as agents in an urban model rather than only as respondents to a questionnaire. None of it is done; the release is the part that had to come first.

10. Conclusions

We have described an openly released set of synthetic populations for 245 Italian municipalities — an entire region above 3 000 inhabitants, 95.1% of its residents — generated from public data alone: aggregate tables for the demographic core, which uses no donor population, and public-use microdata for the two layers that need individual records — no restricted source, no input that a reader cannot obtain. Individuals occupy a census section, a year of age and an address; households satisfy the census size distribution section by section; each attribute is attached at the resolution of the source that determines it, and that resolution is declared. All thirty-seven sources are certified in an executable registry, and the whole release regenerates from a tagged commit, byte for byte.

What the release claims, it measures. Every figure in this paper is taken at that tag against a declared comparison — a census table, a null model, a sampling floor — and where the comparison shows a limit, the limit is reported with its mechanism rather than absorbed. The reader is not asked to trust the pipeline but to run it, and not asked to trust the numbers but to open the address beneath them.

There is an older discipline behind the choice of method. Given macroscopic constraints and nothing else, take the distribution that assumes nothing further and sample the microstates it allows: official statistics publish the constraints, and a population is one of the microstates compatible with them.

Data and code availability

The populations are released as an open dataset under CC-BY-4.0, archived at 10.5281/zenodo.22647404 for the tag `release-v2.0`; 10.5281/zenodo.22647403 resolves to the latest version. The archive is the citable object: single municipalities can be downloaded from the viewer, but the dataset as a whole is distributed only from it. The release is frozen — its files will not be regenerated except to correct an error — and further attribute layers, when they come, will be deposited as new versions of the same record, joined to the population by individual identifier as the household layer already is, leaving every file of this version unchanged.

Two repositories produce and present it, under the MIT licence and archived at the same tag: the generation pipeline (<https://github.com/mirko-degli-esposti/gsp>, 10.5281/zenodo.22646242) and the viewer (<https://github.com/mirko-degli-esposti/animarium>, 10.5281/zenodo.22646253). Regenerating the release requires the pipeline at that tag and the sources listed in its registry, all retrievable as described in Section 3, the survey microdata included. Which municipalities belong to the release is declared in the pipeline too: a registry records each one with the state that admits it, and the deposit is what that registry admits. It carries, for every municipality, the report the pipeline writes, and `SHA256SUMS.txt`, the fingerprint of every released file — the verification path for the reproducibility claim of Section 6.

One source is deliberately outside the deposit. The service locations of Section 8 come from OpenStreetMap, under the Open Database Licence, whose share-alike terms differ from those of everything released here, and whose completeness nobody certifies. The queries, the retrieved points with their timestamps, and the scripts that compare them with a municipal register are in the pipeline repository. They are reproducible; their input is not, since OpenStreetMap on another day is a different dataset — the second reason for keeping it out of a deposit whose point is that it does not change.

The viewer is served at <https://animarium.it>, where the addresses cited in this paper resolve under the commitment of Section 7. The technical report documenting the system and its first twelve municipalities is arXiv:2608.27111.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to draft and revise passages of the manuscript, to assist with the literature search, and to check quantitative statements in the text against the project's own logs and source documents. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. Every reference was verified against the original source, and every quantitative claim in this paper is produced by the deposited code.

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